Composing Trajectory Evaluation with Trust Evaluation

Extracted from a machine-checked development

Abstract

Trajectory systems and trust systems grade claims in different chains. We define a seven-level trust chain and a monotone map ψ from the five-level trajectory chain into it, sending bottoms to bottoms. Composing ψ with canonical trust on the circular and honest trajectories yields concrete transport results: circular evaluation maps entirely to unchecked, while honest evaluation maps the verify event to proved and the claim event to computed. The proofs of all stated results have been machine-checked; we record only the mathematical content.

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1 Introduction

A trajectory system grades agent actions in a five-level chain specialised to verification workflows. A trust system grades mathematical claims in a (typically longer) chain of check strengths used by research agents. To compose the two evaluations—to read a trajectory grade as a trust grade—one needs a map of chains that preserves order and bottom.

In the general theory of trust systems, morphisms transport canonical trust laxly: if ϕ is a monotone map of chains with $\phi(\bot) = \bot$, and F maps claims and steps coherently, then

$$\phi(\tau(x)) \;\leq\; \tau'(F(x)). \tag{1}$$

The present note does not re-prove that abstract transport lemma. Instead it exhibits a concrete chain map ψ from trajectory levels to trust levels and evaluates $\psi \circ \tau$ on the two running examples of the companion note on trajectories: the circular trajectory (where $\tau \equiv \bot$) and the honest trajectory (where τ attains independent grades along sound paths).

The material is elementary. Proofs are indicated by the phrase “by lean,” meaning they have been checked by a proof assistant rather than written out in prose.

What this note claims. In ordinary mathematical language, the claim is: there is a bottom-preserving monotone map ψ from the trajectory chain to the seven-level trust chain such that

- every event of the circular trajectory evaluates, after ψ, to unchecked;
- the honest verify event evaluates to proved, and the honest claim event to computed.

Trajectory evaluation therefore composes with trust evaluation by pure postcomposition of grades; the numerical content of transport is completely explicit on the examples.

2 Recollections

We recall the trajectory chain and the two systems from the companion note; full definitions of trust systems and canonical trust appear there and in the note on trust systems.

Definition 2.1 (Trajectory chain)**.** The **trajectory chain** T_{traj} is

$$\mathsf{unsupported} < \mathsf{shallow} < \mathsf{independent} < \mathsf{crossVerified} < \mathsf{mechChecked}, \tag{2}$$

with bottom unsupported.

Definition 2.2 (Events and systems)**.** Let $E = \{e_0, e_1, e_2, e_3, e_4\}$. The **circular trajectory** $\mathcal{S}_{\mathrm{circ}}$ and the **honest trajectory** $\mathcal{S}_{\mathrm{hon}}$ are trust systems on E graded in T_{traj}, sharing the dependency topology

$$e_0 \to e_1 \to e_2 \to e_3, \qquad e_2 \to e_4, \tag{3}$$

and differing only in grades. Writing τ_{circ} and τ_{hon} for their canonical trusts, the companion note establishes

$$\tau_{\mathrm{circ}}(e) = \mathsf{unsupported} \quad \text{for all } e \in E, \tag{4}$$
$$\tau_{\mathrm{hon}}(e_3) = \mathsf{shallow}, \tag{5}$$
$$\tau_{\mathrm{hon}}(e_4) = \mathsf{independent}. \tag{6}$$

(Along the honest spine e_0, e_1, e_2 one likewise has $\tau_{\mathrm{hon}} = \mathsf{independent}$.)

3 The trust chain

Definition 3.1 (Trust levels)**.** The **trust chain** T_{trust} is the seven-element chain

$$\begin{aligned} \mathsf{unchecked} < \mathsf{speculative} < \mathsf{computed} < \mathsf{proved} \\ < \mathsf{peerReviewed} < \mathsf{published} < \mathsf{leanVerified}, \end{aligned} \tag{7}$$

with bottom $\bot = \mathsf{unchecked}$.

Proposition 3.2 (Chain structure)**.** *T_{trust} carries the structure of a bounded chain: $\leq$ is a linear order, $\bot$ is least, and $\leq$ is decidable.*

Proof. By lean. □

Remark 3.3. Informally these levels grade mathematical claims in a research-agent deployment: from no check, through speculative and computed evidence, to proved, peer-reviewed, published, and machine-verified claims. The precise operational meaning of each level is not needed for the theorems below; only the order and the bottom element matter.

4 The trajectory–trust map

Definition 4.1 (The map ψ)**.** Define $\psi : T_{\text{traj}} \to T_{\text{trust}}$ by

Trajectory level	Trust level
unsupported	unchecked
shallow	computed
independent	proved
crossVerified	peerReviewed
mechChecked	leanVerified

Remark 4.2. The map skips speculative and published: trajectory levels are coarser than trust levels, and ψ lands on the trust grades that best match the intended strength of each trajectory grade. In particular, independent trajectory checks are read as proved claims, and mechanically checked trajectory steps as lean-verified claims.

Theorem 4.3 (Bottom preservation)**.**

$$\psi(\bot_{T_{\text{traj}}}) \;=\; \bot_{T_{\text{trust}}}. \tag{8}$$

Equivalently, $\psi(\mathsf{unsupported}) = \mathsf{unchecked}$.

Proof. By lean. □

Theorem 4.4 (Monotonicity)**.** *For all* $a, b \in T_{\text{traj}}$*,*

$$a \le b \implies \psi(a) \le \psi(b). \tag{9}$$

Proof. By lean. □

Remark 4.5 (Chain-morphism requirements)**.** In the abstract theory of morphisms of trust systems, a map of check-posets must be monotone and strict at bottom. Theorems 4.3 and 4.4 record exactly those two requirements for ψ. Together they make ψ eligible as the grade component of a trust-system morphism from a trajectory system into a trust system graded in T_{trust}.

5 Concrete transport

Composing ψ with canonical trust on the two examples yields fully evaluated trust grades—no inequality, only equality.

Theorem 5.1 (Circular trajectory transports to unchecked)**.** *For every event* $e \in E$*,*

$$\psi(\tau_{\text{circ}}(e)) \;=\; \mathsf{unchecked}. \tag{10}$$

Proof. By lean. □

Remark 5.2. This is immediate from circular collapse ($\tau_{\text{circ}} \equiv \mathsf{unsupported}$) and bottom preservation of ψ. It is recorded as a theorem because it is the operational statement used by deployments: after transport, a circular trajectory contributes no trust above unchecked at any event.

Theorem 5.3 (Honest verify event)**.**

$$\psi(\tau_{\text{hon}}(e_4)) \;=\; \mathsf{proved}. \tag{11}$$

Proof. By lean. □

Theorem 5.4 (Honest claim event)**.**

$$\psi(\tau_{\mathrm{hon}}(e_3)) \;=\; \mathsf{computed}. \tag{12}$$

Proof. By lean. □

Remark 5.5 (Reading the honest results)**.** On the honest trajectory, $\tau(e_4) = \mathsf{independent}$ and $\tau(e_3) = \mathsf{shallow}$. Applying ψ sends independent to proved and shallow to computed. Thus a sound verify branch is trusted at the proved level after composition, while a shallow claim branch is trusted only at the computed level—matching the intended gap between verification and assertion.

6 Discussion

What has been established. Trajectory grades embed into trust grades via a bottom-preserving monotone map ψ. On the circular example the composite is constantly unchecked; on the honest example the verify and claim events map to proved and computed respectively. Composition of the two evaluations is therefore a matter of postcomposing the grade map—no further analysis of dependency structure is required once τ is known.

Relation to abstract transport. The general transport lemma for morphisms of trust systems asserts a *lax* inequality $\phi(\tau(x)) \leq \tau'(F(x))$. The results here are stronger on the examples because they evaluate $\psi \circ \tau$ exactly, without constructing a target system S' or a claim map F. They may be read as the grade half of a morphism, specialised to the case where one only needs the image of source canonical trust under ψ.

Scope. The development treats a single map of chains and two finite examples. It does not construct a full morphism of trust systems (claim and step components), nor does it address naturality of transport under composition of morphisms. Those belong to the general theory.

Machine checking. Every definition and theorem statement above corresponds to a declaration in a machine-checked formalisation. The phrase "by lean" indicates that the proof has been verified by a proof assistant; we have not reproduced the formal proof scripts here. A mathematician reading this note need only verify that the English statements match the intended mathematics; the formal system is responsible for the derivations.

Acknowledgements

The mathematical content composes the trajectory systems of the companion note with the chain-morphism requirements of the theory of trust systems. The presentation here is a prose rendering of a formal development intended for human review of the theorem statements.

References

[1] Trust systems: the certificate lemma and transport of trust. Companion formal development.

[2] Trajectory systems: circular verification collapses canonical trust. Companion note.